ExtremaFuse for Adaptive Compression of Frame-Level Speech Representations

Anonymous ACL submission

Abstract

How much can a recognizer shorten a speech sequence while retaining the distinctions needed to recognize phones? ExtremaFuse selects fusion centers at peaks and valleys of the cosine distance between adjacent frames and learns how to weight the five frames around each center. We apply it to Mel features and again inside an offline Conformer transducer trained on 460 hours of LibriSpeech. On test-clean, this system has a phone error rate of 5.69 percent, compared with 5.74 percent for a stride-four baseline with the same input projection width and a wider encoder. Relative to fusion at the Mel input alone, the second stage removes 59.8 percent of the remaining steps at a cost of 0.11 percentage points in phone error. On the full test-clean set, the two-stage system uses 74.9 percent fewer major multiply-accumulate operations per second of input audio, including the predictor and joiner. Measured fusion weights show how neighboring frames still contribute to the shorter sequence.

1 Introduction

A stride-four encoder represents every four input frames with one time step, whether those frames cover a steady vowel or straddle a transition. A phone recognizer offers a direct test of this choice. A system might emit a shorter sequence yet introduce deletions where a brief sound becomes hard to distinguish. The arithmetic cost also depends on which encoder layers process the shortened sequence and how many candidate hypotheses the decoder scores.

We propose ExtremaFuse, a local operation on a sequence of frame vectors. Peaks and valleys in adjacent-frame distance determine where to emit a vector. The five frames around each selected position determine its value through learned weights. The operator can act on acoustic features or intermediate encoder states and takes no phone boundaries as input.

We ask two questions. First, within the computation budget of a fixed-stride recognizer, what phone error rate can the adaptive system achieve? We compare systems with the same per-frame projection width and count major multiply–accumulate operations along the actual decoding path. Second, what does an additional round of compression cost after the Mel sequence has already been shortened? We count the steps and phone errors, inspect which input frames contribute to an output, and test a length-matched replacement in the same trained model. A breakdown by speaking rate reveals which edit types account for the difference in aggregate error.

2 Related Work

Time reduction for speech models. Funnel pooling reduces the frame rate inside large speech encoders, and fixed-stride subsampling has been studied at both the input and the output of Whisper (Wang et al., 2024; Cho et al., 2026). Adaptive methods also exploit neighboring representations. A-ToMe merges adjacent transducer tokens based on the similarity of their attention keys (Li et al., 2023). In audio language models, Bhati et al. (2025) compare segmentation from adjacent-frame distances with uniform pooling, and Xiang et al. (2026) study similarity-based merging at multiple depths. ExtremaFuse uses extrema in the distance curve as centers for overlapping, learned windows. This design lets us inspect which frames actually contribute to each output.

Local acoustic change and phone evaluation. Changes between neighboring features can provide cues for phonetic segmentation (Kreuk et al., 2020; Yantosca and Cheng, 2025). Here the selected positions serve as fusion centers. Their value is assessed through recognition of the reference phone sequence. Work on phone realization illustrates the range of distinctions hidden by a sin-

Input	$H = (h_1, \ldots, h_T)$
Select	$H \longmapsto (d_1, \ldots, d_{T-1}) \longmapsto C$
Fuse	$(H, C) \longmapsto (z_{c_1}, \ldots, z_{c_K})$

Figure 1: ExtremaFuse at one insertion site. Distances determine an ordered set C of centers. Each z_c pools the original features around that center with learned weights. The next stage receives K frames of dimension D.

gle aggregate score (Bharadwaj et al., 2026). We therefore also examine substitutions, deletions and insertions.

3 ExtremaFuse

3.1 Sequence interface

The input is an ordered sequence $H = (h_1, \ldots, h_T)$ of valid D-dimensional frames. The output has the same feature dimension and a content-dependent length $K \leq T$. Downstream layers consume the new length and mask. Figure 1 shows how the original frames enter both selection and fusion.

3.2 Selecting fusion centers

For adjacent nonzero frames, define $d_t = 1 - \cos(h_t, h_{t+1})$. A peak at p selects frame $p + 1$, immediately after a locally large change in direction. A valley at v selects frame v at a locally small change. We merge duplicate positions and keep them in chronological order. If a nonempty sequence has no interior extremum, we use its first frame. The number of centers follows from the input, without a target retention rate.

3.3 Learning local fusion weights

For a center c, let W_c contain its valid neighbors up to two positions away, including c. A two-layer network takes the five concatenated frame vectors, applies a 32-unit ReLU layer and predicts five logits. Inputs outside the sequence use the nearest endpoint for this prediction. The softmax weights are renormalized over valid positions for the output,

$$z_c = \sum_{j \in W_c} \alpha_{cj} h_j, \quad \alpha_{cj} \geq 0, \quad \sum_{j \in W_c} \alpha_{cj} = 1. \tag{1}$$

Overlapping windows allow one input to contribute to more than one output. The center-selection rule has no trainable parameters. Each insertion site learns its own fusion weights through the recognition objective.

4 Experimental Setup

4.1 Task and systems

We train on LibriSpeech train-clean-100 and train-clean-360 (460 hours) and recognize sequences from a 69-phone ARPABET inventory with vowel stress (Panayotov et al., 2015). We evaluate ExtremaFuse in a ten-layer offline Conformer with a transducer objective (Gulati et al., 2020; Graves, 2012). In the two-stage system (E06), fusion first shortens the 80-dimensional Mel sequence before its input projection. A second module acts after layer 5, so the final five encoder layers receive the twice-compressed sequence.

Four-frame stacking couples the input projection and encoder width in the baseline. E05 matches the adaptive systems' 128-dimensional per-frame projection and stacks four projected vectors, giving it a 512-dimensional encoder. The adaptive systems use stride one and a 128-dimensional encoder (Table 1). We train with seed 42 and select checkpoints by dev-clean validation loss. Batch budgets and update counts differ across runs. Appendix A records the updates at selection and at the end of training.

4.2 Uniform pooling control

Uniform pooling tests whether comparable compression at the same sites suffices. The planned control uses E06's backbone and replaces both fusion modules with means over contiguous, disjoint segments of as nearly equal length as possible. At each site, its output length depends only on its input length and a retention ratio calibrated from E06 on dev-clean. The rest of the recognizer will be trained from scratch with these two ratios fixed.

4.3 Quality, cost, and uncertainty

Corpus phone error rate (PER) divides total substitutions, deletions and insertions by the number of reference phones. We use the same 2,620 test-clean utterances for every system (189,023 phones, 40 speakers). Paired intervals resample speakers jointly across systems and recompute corpus PER on each of 10,000 draws. Test-clean informed the choice of configurations, so comparisons on this split are exploratory.

We measure computation by major multiply–accumulate operations (MACs) per second of in-

System	Input proj.	Encoder width	Params (M)	test-clean PER (%)	test-clean GMAC/s	test-other PER (%)	test-other GMAC/s
Fixed stride 4 (E05)	128	512	29.58	5.7427	0.92697	–	–
ExtremaFuse, Mel + layer 5 (E06)	128	128	4.25	5.6882	0.23252	–	–
ExtremaFuse, Mel only (E02)	128	128	4.23	5.5782	0.38207	–	–
ExtremaFuse, layer 2 (E01)	128	128	4.24	5.5141	0.48637	–	–
ExtremaFuse, layer 5 (E03)	128	128	4.24	4.7809	0.62512	–	–
ExtremaFuse, layer 7 (E04)	128	128	4.24	4.8603	0.70362	–	–
Uniform pooling, Mel + layer 5	128	128	–	–	–	–	–

Table 1: Phone recognition quality and major MACs per second of input audio (GMAC/s), including encoder, compression, predictor and joiner operations. All six systems use a 128-dimensional per-frame input projection. Both metrics are lower-is-better. Dashes mark the planned pooling control and test-other measurements.

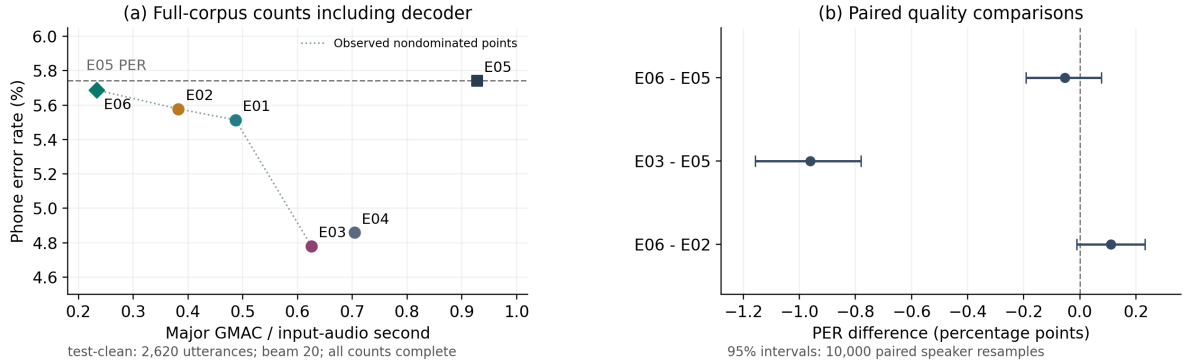


Figure 2: Test-clean PER and full-corpus computation including the decoder for the six retained systems (left). The horizontal line marks E05's PER; the dotted line joins observed nondominated points without estimating intermediate performance. Paired PER differences in percentage points (right) use 10,000 paired speaker resamples.

put audio. For utterance counts C_i and actual audio durations d_i, the reported value is $\sum_i C_i/(10^9 \sum_i d_i)$. One MAC denotes one multiply–accumulate. We use FP32, batch size one and beam width 20. Counts follow each layer's actual sequence length and the hypotheses expanded by beam search, with every output checked against the archived prediction.

The scope includes linear and convolution operations, both attention matrix products, fusion networks, adjacent-frame dot products, weighted fusion, predictor gates and joiner projections. We separate encoder and compression costs from predictor and joiner costs. Feature extraction, normalization, softmax, activations, extrema comparisons, sorting and memory access are outside this major-MAC scope. For the position study, a candidate meets the computation budget when its count divided by E05's count on the same split and decoding protocol is at most one.

5 Results

5.1 Recognition and computation

The two-stage system (E06) has a test-clean PER of 5.6882%, compared with 5.7427% for E05. E06 makes 103 fewer errors, a difference of -0.054 percentage points with a paired 95% interval of $[-0.192, +0.076]$. The interval leaves the quality ordering uncertain. E06 uses 0.23252 GMAC per second of input audio, compared with 0.92697 for E05, a reduction of 74.92%. Their respective encoder costs, including compression and projections, are 0.13207 and 0.80207. Predictor and joiner costs are 0.10045 and 0.12491. The systems also differ in encoder width.

E03 provides a higher-quality operating point at 4.7809% PER. Relative to E05, its PER difference is -0.962 percentage points, with a paired 95% interval of $[-1.157, -0.781]$. Its full-corpus count is 0.62512 GMAC per audio second, or 67.44% of the baseline count (Figure 2).

The completed counts also distinguish encoder savings from candidate-scoring work. Moving single-stage fusion from layer 5 (E03) to layer 2

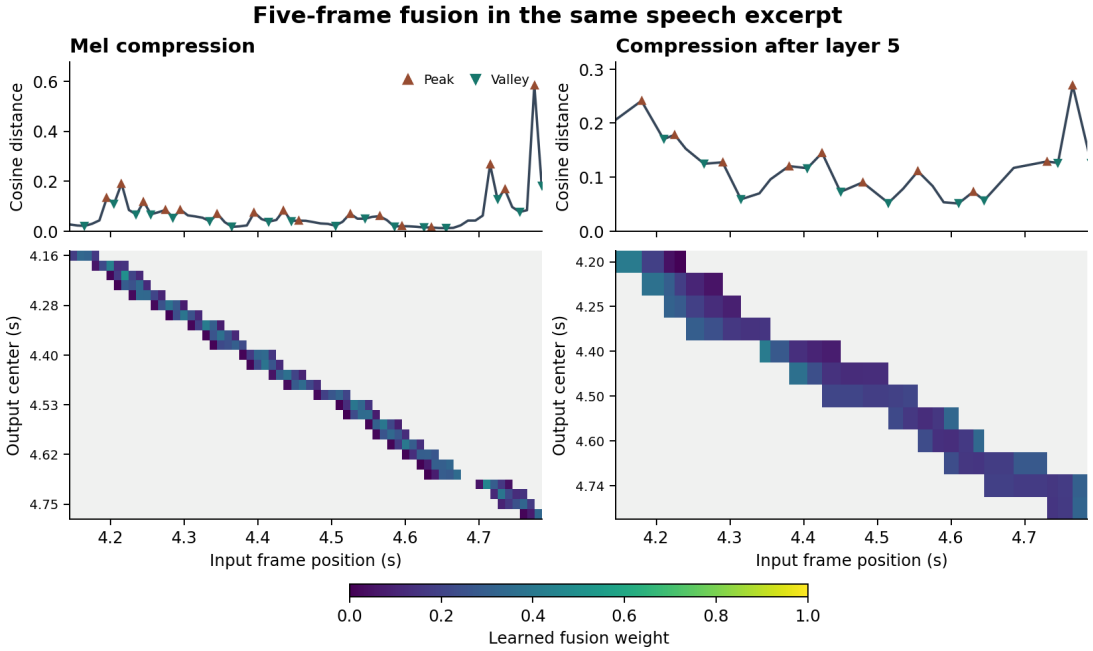


Figure 3: Measured distance curves and fusion weights for the same excerpt at the Mel input and after layer 5. Each row in the lower panels represents an output center. Layer-5 positions are expressed on the Mel time axis; their features already incorporate encoder context.

(E01) reduces encoder and compression MACs from 0.43193 to 0.29744 GMAC per audio second, a 31.14% reduction. Predictor and joiner costs are 0.19319 and 0.18893, respectively, so the total falls by 22.20%, from 0.62512 to 0.48637. PER increases from 4.7809% to 5.5141%. Layer-7 fusion (E04) uses 0.70362 GMAC per audio second at 4.8603% PER; its point estimate improves neither metric over layer 5. All five adaptive systems fall below E05's major-MAC budget. Mel-only fusion (E02), the direct front-end replacement, uses 0.38207 total GMAC per audio second at 5.5782% PER. These are discrete trained systems; connecting the observed nondominated points does not establish performance between them or a multi-budget curve for fixed stride.

5.2 Independent evaluation and pooling

Two comparisons remain open. On test-other we will test whether E06 stays within a prespecified acceptable PER increase over E05 and report the corresponding full-corpus MAC ratio. We will also compare E06 with independently trained uniform pooling at the same two sites and report realized frame counts, PER and major MACs. The empty entries in Table 1 are reserved for these measurements.

6 Analysis

6.1 The cost of the second fusion stage

To examine the second compression itself, the relevant reference is the Mel-only system (E02). Across test-clean, both E02 and E06 produce 764,950 Mel-stage outputs. E06 reduces this to 307,404 steps after layer 5, removing a further 59.81%. The final sequence retains 15.79% of the original Mel frames. Computation including the decoder falls from 0.38207 to 0.23252 GMAC per audio second, a reduction of 39.14%.

E06 makes 208 more errors than E02. PER increases from 5.5782% to 5.6882%, a difference of $+0.110$ percentage points (paired 95% interval $[-0.011, +0.232]$). These are separately trained systems. Appendix B.1 extends this comparison by moving the second Mel-family stage to layer 3 or 7, while retaining the completed Mel-plus-layer-5 system.

6.2 Window coverage and a matched-length intervention

Equation 1 gives a bound on local distortion,

$$\|z_c - h_c\|_2 \leq \sum_{j \in W_c} \alpha_{cj} \|h_j - h_c\|_2. \qquad (2)$$

When the neighbors receiving high weights resemble the center, the fused vector stays close to it.

Group	Phones	ΔS	ΔD	ΔI	ΔPER
Slow	36,624	+55	−2	−40	+0.035
Middle	83,207	+95	−326	+279	+0.058
Fast	69,192	+85	−504	+255	−0.237

Table 2: E06 minus E05 on test-clean by speaking rate. S, D and I count substitutions, deletions and insertions. PER differences are in percentage points.

This gives a local explanation that can be checked against the actual inputs and weights. In 100 test-clean utterances (ten per speaking-rate decile, 37 speakers), adjacent vectors at layer 5 have a mean cosine similarity of 0.877. Pairs separated by half an utterance under a circular shift have a mean similarity of 0.143.

On this sample the second stage emits 40.19% as many vectors as it receives. Its windows cover 98.24% of the input positions, and 97.03% receive a weight of at least 0.05 in at least one output. The effective number of inputs per output, $1/\sum_j \alpha_{cj}^2$, averages 3.89. Figure 3 shows how a frame between two centers can contribute to both. Coverage measures participation in fusion. Phone errors quantify which distinctions survive recognition.

Could an ordinary mean produce a similar result once the encoder has supplied context? We freeze E06, keep its Mel module, and replace only the layer-5 module with means over equally spaced segments. For every utterance we use exactly the number of outputs that E06 selected. Reference-transcript negative log-likelihood changes from 0.1423 to 0.1435 nats per phone (difference $+0.00117$, paired 95% interval $[-0.00241, +0.00499]$). Thus the frozen model is also compatible with simple local averaging on this sample. The independently trained two-site control in Table 1 tests whether adaptive fusion improves recognition or decoding cost at a similar frame budget.

6.3 Speaking rate and the error profile

The small aggregate PER difference hides a change in error type. Across test-clean, E06 makes 832 fewer deletions than E05, alongside 235 more substitutions and 494 more insertions. To locate these changes, we group utterances by reference phones per minute of audio, including pauses. The slow and fast groups each contain three deciles and the middle group contains four.

The fast group accounts for 164 fewer errors, while the slow and middle groups contribute 13 and 48 additional errors (Table 2). E06 retains similar proportions of the original Mel frames across the three groups (15.74%, 15.77% and 15.84%). The fast-group PER difference is -0.237 percentage points (95% interval $[-0.447, -0.010]$). Testing whether the benefit is larger in fast speech requires comparing groups directly. The fast-minus-slow difference is -0.273 percentage points with an interval of $[-0.697, +0.154]$, leaving the rate dependence uncertain.

7 Conclusion

On LibriSpeech test-clean, applying ExtremaFuse at the Mel input and inside the encoder gives a phone error rate close to the stride-four baseline. The second stage cuts the remaining sequence by more than half. Its overlapping windows still assign weight to most input positions, and the frozen model tolerates replacing them with length-matched averages on the diagnostic sample. The completed counts show that moving fusion earlier reduces encoder work while candidate scoring limits the total reduction. E06 uses 25.08% of E05's major MACs on this split. An independently trained pooling control is needed to test the contribution of adaptive fusion at a comparable frame budget.

Limitations

Our evidence covers English read speech, phone recognition, one offline Conformer transducer and one training seed. The fixed-stride and adaptive systems differ in capacity or training exposure. Major MACs cover the disclosed arithmetic scope and depend on the input lengths and beam-search path. They omit scalar operations and memory traffic and do not establish wall-clock performance. The speaker intervals quantify test-sample uncertainty for these checkpoints. Test-clean was used to develop the comparisons, and the reference transcripts supply no aligned phone boundaries with which to score the selected centers.

Ethical Considerations

The evaluation uses English audiobook recordings and stressed phone labels. Application to other speaking styles or accents would require corresponding evaluation data.

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A Configurations and Reproducibility

Table 3 gives the configurations behind Figure 2. For E05 the validation loss selected an earlier checkpoint, at 23,150 of 117,500 updates. E06 was resumed with an increased phone-token batch budget and selected at 89,901 of 90,986 updates. Test scores throughout the paper correspond to the selected checkpoints.

All systems use 16 kHz audio, 80 Mel bins and a 10 ms frame hop. Training uses Adam with a peak learning rate of 0.0005, 10,000 warm-up updates and gradient clipping at 10. The corpus bootstrap resamples 40 speakers with replacement using seed 42 and reports percentile intervals. All utterances from a sampled speaker enter each system's numerator and denominator together.

B Additional Diagnostics

B.1 Testing early placements under the computation budget

The stride-one, layer-2-only system (E01) uses 0.48637 GMAC per audio second on test-clean, or 52.47% of E05's total. We extend the placement study with the prespecified grid in Table 4. Completed placements are reused rather than trained again. New runs move the fusion sites in the corresponding reference recipe, preserving the backbone, fusion rule, optimization settings, seed and 50-epoch target. They use independently calibrated phone-token batch budgets; actual update counts and validation histories will therefore be reported rather than described as matched optimization exposure.

All supplementary adaptive runs use stride one and a 128-dimensional projection and encoder.

ID	Compression site(s)	Stride	Proj./width	Selected step	Final step	PER (%)	GMAC/s
E01	Layer 2	1	128/128	69,137	126,550	5.5141	0.48637
E02	Mel	1	128/128	121,688	126,550	5.5782	0.38207
E03	Layer 5	1	128/128	151,700	165,000	4.7809	0.62512
E04	Layer 7	1	128/128	118,826	126,550	4.8603	0.70362
E05	Fixed stride	4	128/512	23,150	117,500	5.7427	0.92697
E06	Mel + layer 5	1	128/128	89,901	90,986	5.6882	0.23252

Table 3: Full test-clean PER and major MACs per second of input audio, including the decoder. All adaptive configurations use learned five-frame fusion. Stride is the fixed stacking factor.

Family	Fusion site(s)	Status
One stage	Layers 1, 3, 9	Pending
One stage	Layers 5, 7	Reuse E03, E04
Two stages	Mel + layers 3, 7	Pending
Two stages	Mel + layer 5	Reuse E06
Two stages	Layers 2 + 7	Pending
Two stages	Layers 2 + 5	Reuse archive E08
Reference	Layer 2; Mel only	Reuse E01, E02

Table 4: Supplementary placement grid. Layer numbers identify global Conformer outputs; "Mel" is the input-feature site. Pending runs have no reported scores. The layers-2-plus-5/7 family uses a hard three-frame neighborhood at its first stage and learned five-frame fusion at its second stage. Single-stage and Mel-plus-layer runs use learned five-frame fusion. Archive IDs remain distinct from paper IDs.

We will report stage input and output counts, corpus PER and full-corpus major MACs, including any over-budget result. The historical layers-2-plus-5 reference retains its recorded batch-resume history and its hard-first, learned-second fusion rules, and is kept separate from the Mel family. Candidate selection uses the E05-relative computation budget on dev-clean before confirmation on an independent split.

B.2 Extending the natural speaking-rate comparison

We reuse the frozen test-clean decile membership for every completed system: deciles 1–3 are slow, 4–7 middle and 8–10 fast, containing 786, 1,048 and 786 utterances respectively. The labels describe natural reference-phone rate over the full recording, including pauses; no audio is time-stretched. Within a group, PER divides total phone errors by total reference phones, and major GMAC per audio second divides total counted operations by total audio duration. Encoder/compression, predictor and joiner costs remain separate. Existing predictions and completed per-utterance counts are aggregated directly; completed systems are neither retrained nor decoded again for this analysis.

Table 5 reports the ten already completed systems. The supplementary layer grid will enter these same groups after its full-corpus evaluations are verified. These test-clean comparisons are descriptive and conditional on the selected checkpoints. New paired uncertainty intervals and a fast-minus-slow interaction contrast remain to be computed; independent confirmation requires thresholds frozen on dev-clean. Point estimates alone do not establish a rate-specific advantage.

B.3 Selection of the illustrated excerpt

For the weight diagnostic we select ten utterances per test-clean speaking-rate decile using a deterministic hash of the utterance identifier. Figure 3 shows a 0.64-second window centered at the midpoint of the lowest-hash eligible utterance in decile five (at least 200 raw frames). The choice therefore does not use recognition errors or the appearance of the weights.

System	Slow		Middle		Fast	
	PER (%)	GMAC/s	PER (%)	GMAC/s	PER (%)	GMAC/s
Layer 2 (E01)	5.5401	0.470140	5.2628	0.491895	5.8027	0.491566
Mel only (E02)	5.8459	0.367544	5.3505	0.384496	5.7102	0.390297
Layer 5 (E03)	4.8793	0.602607	4.6066	0.634487	4.9384	0.629924
Layer 7 (E04)	5.0568	0.677281	4.6631	0.716562	4.9934	0.706431
Stride4 (E05)	5.9142	0.917053	5.3469	0.930294	6.1279	0.930231
Mel + layer 5 (E06)	5.9497	0.222101	5.4046	0.233596	5.8909	0.239352
Layers 2 + 5 (archive E08)	5.5319	0.545851	5.1859	0.569498	5.6452	0.565460
Stride1, projection128	5.1524	0.986788	4.8217	1.055389	5.0945	1.054391
Stride2, projection128	5.3490	0.838819	4.6306	0.868587	4.9832	0.869017
Stride8, projection128	5.4554	1.303961	6.0728	1.313081	8.0544	1.316922

Table 5: Natural speaking-rate groups on the frozen full test-clean set. Each group reports corpus PER and total major GMAC per second of its input audio, including the actual beam-20 predictor/joiner path. Slow/middle/fast contain 786/1,048/786 utterances. All ten systems reuse completed predictions and full counts; no additional training or decoding was performed for this table. Archive E08 uses hard three-frame neighborhoods first and learned five-frame fusion second. These are descriptive point estimates for fixed checkpoints with different training histories and capacities, not causal estimates of layer or rate effects.